

Variables in an Experiment

Change one variable, measure one variable, control the rest. That is what makes a test fair.

Name _____ Date _____

Worked example

Question: Does ramp height change how far a toy car rolls?

Independent: ramp height (cm).

Dependent: distance rolled (cm), measured with a ruler.

Controlled: same car, same ramp surface, same starting line.

My investigation question

Independent variable The one thing you change on purpose.

Dependent variable The thing you measure. Give the units.

Controlled variables Everything you deliberately keep the same.

Is my test fair? If someone changed something else as well, could you still trust the result?
